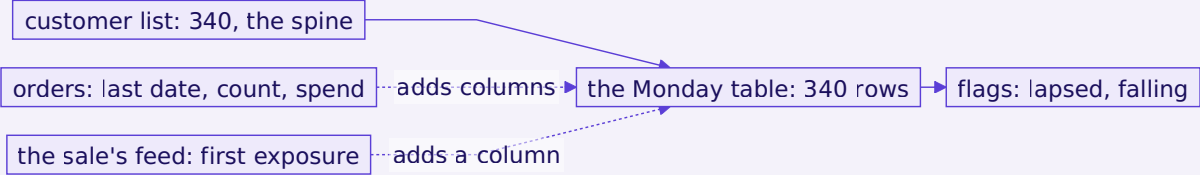


Can the growth team act on Monday's table without checking it?

Kalpa Retail's growth team wants one row per customer, rebuilt every Monday in pandas from the warehouse: recency, frequency, spend, segment, the monsoon sale's reach and the flags it acts on.

1 Which source decides the table's rows, and what does each source add?



The customer list decides the rows; the orders and the campaign feed only add columns. Every Monday the table agrees with three numbers before it leaves: 340 rows, Rs 19,84,00,000 of spend, and the as-of date, 28 September 2026, the last date the data covers.

Crux. Start a customer table from the customer list, because a table built from orders leaves out everyone who never ordered: 39 of Kalpa's 340.

2 Which way builds one row per customer, and when does it move to the warehouse?

Option	Rows moved	Lines	Best when
Week 1's loop	1,000	6	Explaining one case line by line
SQL GROUP BY	301	7	The orders run to crores
pandas groupby	1,000	3	The next steps need the orders in memory

```
rfm = orders.groupby("customer_id").agg(
    last_order=("order_date", "max"),
    frequency=("order_id", "count"),
    spend=("amount", "sum")).reset_index()
```

Merge it onto the customer list, then fill a missing count with 0 on purpose: a missing value is never equal to 0.

3 How does a merge keep the table at one row per customer?

merge with no how is an inner join. validate="one_to_one" promises each key at most once on each side and raises MergeError before a wrong table exists. A customer sent twice is a business question: the growth team keeps the first exposure.

```
first = (feed.sort_values("exposed_date")
        .drop_duplicates("customer_id", keep="first"))
table = table.merge(first, on="customer_id",
                   how="left", validate="one_to_one")
```

Crux. Write how= and validate= on every merge, because one repeated key adds a customer's whole spend again and validate stops the merge before the table exists.

4 What does pivot_table put in each cell, and which shape answers which question?

Shape	One row is	Answers
Long, groupby on two keys	A member and a month	A trend, month by month
Wide, pivot_table	A member, a column per month	A comparison along the row
Wide back to long, melt	A member and a month, zeros kept	A chart of every month

Crux. Write aggfunc= on every pivot and check its grand total against the source, because pivot_table averages by default and turned a 29 percent fall into 18.

5 Why did three tools disagree about who bought, and how did they come to agree?

Tool	A customer with no segment
Plain Python dictionary	Filed under None
SQL GROUP BY	One NULL group
pandas groupby	Dropped, unless dropna=False

The 23 reached customers who never ordered had no segment in their orders, so pandas said 100 percent. With the segment from the customer list, all three say 107 of 130, 82 percent. The groups must add back to the rows.

6 Which tool owns which recurring number?

The ask	Owner	Why
Finance's revenue	SQL	Finance reruns it where the data lives; it moves 8 rows
The customer table	pandas, reading the warehouse	Analysts add columns weekly
The months view	pandas	It pivots data in memory
An auditor's one-off	Plain Python	Each step is a readable line one-off

A pandas notebook is refused for Finance. Size by rows moved: SQL 8, pandas 1,340.

Crux. Two tools agree only when they share one definition, so give each recurring number one owner, chosen by who reruns it and sized by the rows each route moves.

7 What makes a Monday refresh safe to leave alone?

Recency counts to orders["order_date"].max(), carried as as_of. Counted to 19 October, the win-back list read 166 where the data holds 111; the 21 days between are the data's age.

Guard	Catches
One row per customer	A repeated key
Rows equal the customer list	A customer missing or doubled
Spend equals the warehouse	A customer's spend counted twice
Smallest recency is 0	Recency counted to the wrong day

Crux. Count recency to the data's own last date and let the refresh refuse a table that fails a guard, because the wall clock put 166 customers on a list that holds 111.

THE WORDS ON THIS SHEET

- RFM** Recency, frequency, monetary: days since last order, order count and spend
- Spine** The table whose rows decide the result's rows
- groupby** Split rows by a key, apply a calculation, combine one row per key
- validate** The promise a merge states about its keys, raising MergeError when broken
- First touch** The growth team's rule: a customer reached twice keeps only the first date
- Merge** pandas' join, in four shapes set by how